

Homework 1

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Problem 1

(a)

$$2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 4]^T$$

$$2u = [4, -2, 6]^T$$

$$2u - v = [4 - 1, -2 - 4, 6 - (-2)]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$$

$$[\text{ANSWER: } (2u - v) \cdot w = 24]$$

(b)

$$(1/2)w = [-2, 1, 3]^T$$

$$v + (1/2)w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$||v + (1/2)w|| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3}$$

$$[\text{ANSWER: } 3\sqrt{3} \approx 5.20]$$

(c)

$$\cos \theta = (u \cdot w) / (||u|| ||w||)$$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$||u|| = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$||w|| = \sqrt{16 + 4 + 36} = \sqrt{56}$$

$$||w|| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$||u|| ||w|| = \sqrt{14} \cdot 2\sqrt{14} = 28$$

$$\cos \theta = 8/28 = 2/7$$

$$\theta = \arccos(2/7)$$

$$[\text{ANSWER: } \theta \approx 73.40^\circ]$$

Problem 2

(a)

AB is undefined, because A is 3×3 (3 columns) and B is 2×3 (2 rows), and $3 \neq 2$.

BA is 2×3 since B is 2×3 and A is 3×3 .

row 1: $[1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(1),$

row 1: $[1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(1), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$

row 2: $[3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$

[ANSWER: $BA = [2 \ -9 \ -2]$

$[7 \ 4 \ -1]]$

(b)

AV is $(3 \times 3)(3 \times 1) = 3 \times 1$.

$AV = [2(1) + (-1)(-2) + 0(3)] \ [\ 4]$

$[1(1) + 3(-2) + (-2)(3)] = [-11]$

$[0(1) + 4(-2) + 1(3)] \ [\ 5]$

VA is undefined: V is 3×1 so it has 1 column, A has 3 rows, and $1 \neq 3$.

[ANSWER: $AV = [4, -11, 5]^T$, VA undefined]

(c)

Aw is undefined: A has 3 columns and w is 1×3 with only 1 row, $3 \neq 1$.

$WA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)]$

[ANSWER: $WA = [4 \ -6 \ 1]$, Aw undefined]

(d)

vw is $(3 \times 1)(1 \times 3) = 3 \times 3$, so each entry is a v entry times a w entry.

$vw = [1(2) \ 1(0) \ 1(-1)] \ [2 \ 0 \ -1]$

$[-2(2) \ -2(0) \ -2(-1)] = [-4 \ 0 \ 2]$

$[3(2) \ 3(0) \ 3(-1)] \ [6 \ 0 \ -3]$

ww is $(1 \times 3)(3 \times 1) = 1 \times 1$ and equals the dot product: $2(1) + 0(-2) + (-1)(3) = -1$

[ANSWER: w as above, $wv = [-1] \]$

(e)

$A^2 = AA$, defined since A is square.

row 1: $[2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$

row 2: $[1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$

row 3: $[0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 12, -7]$

[ANSWER: $A^2 = [3 \ -5 \ 2]$

$[5 \ 0 \ -8]$

$[4 \ 12 \ -7] \]$

(f)

B^2 undefined.

Problem 3

(a)

$$[1 \ 2 \ -1 \mid -1]$$

$$[2 \ 3 \ 1 \mid 2]$$

$$[-1 \ 1 \ 2 \mid -1]$$

$$R_2 \rightarrow R_2 - 2R_1 \quad [1 \ 2 \ -1 \mid -1]$$

$$R_3 \rightarrow R_3 + R_1 \quad [0 \ -1 \ 3 \mid 4]$$

$$[0 \ 3 \ 1 \mid 0]$$

$$R_2 \rightarrow -R_2 \quad [1 \ 2 \ -1 \mid -1]$$

$$R_3 \rightarrow R_3 - 3R_2 \quad [0 \ 1 \ -3 \mid -4]$$

$$[0 \ 0 \ 10 \mid 12]$$

$$R_3 \rightarrow (1/10)R_3 \quad [1 \ 2 \ -1 \mid -1]$$

$$[0 \ 1 \ -3 \mid -4]$$

$$[0 \ 0 \ 1 \mid 6/5]$$

$$x_3 = 6/5$$

$$x_2 = -4 + 3(6/5) = -4 + 18/5 = -2/5$$

$$x_1 = -1 - 2(-2/5) + 6/5 = -1 + 4/5 + 6/5 = 1$$

$$[\text{ANSWER: } x = [1, -2/5, 6/5]^T]$$

(b)

$$R_2 \rightarrow R_2 - R_1$$

$$[1 \ 2 \ -1 \mid 4]$$

$$[2 \ 5 \ 1 \mid 6]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_2 \rightarrow R_2 - 2R_1 \quad [1 \ 2 \ -1 \mid 4]$$

$$R_3 \rightarrow R_3 - 3R_1 \quad [0 \ 1 \ 3 \mid -2]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$R_3 \rightarrow R_3 - R_2 \quad [1 \ 0 \ -7 \mid 8]$$

$$R_1 \rightarrow R_1 - 2R_2 \quad [0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

No pivot in column 3, so z is free. Let $z = t$. Then $y = -2 - 3t$ and $x = 8 + 7t$, so the system has infinitely many solutions.

$$[\text{ANSWER: } [x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}]$$

Problem 4

(a)

Let x_1 = kg of Alloy 1 used, x_2 = kg of Alloy 2 used, x_3 = kg of Alloy 3 used.

The 100 kg product has 36 kg copper, 44 kg zinc, and 20 kg nickel, and no metal is

lost, so for each metal the mass in equals the mass in the product.

copper: $0.5x_1 + 0.2x_2 + 0.4x_3 = 36$

zinc: $0.3x_1 + 0.7x_2 + 0.2x_3 = 44$

nickel: $0.2x_1 + 0.1x_2 + 0.4x_3 = 20$

(b)

Multiply everything by 10:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

(c)

$$R_1 \rightarrow R_1 - 2R_3 \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_2 \rightarrow R_2 - 3R_1 \quad [1 \ 0 \ -4 \mid -40]$$

$$R_3 \rightarrow R_3 - 2R_1 \quad [0 \ 7 \ 14 \mid 560]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2$$

$$R_2 \rightarrow (1/7)R_2 \quad [1 \ 0 \ -4 \mid -40]$$

$$R_3 \rightarrow R_3 - R_2 \quad [0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 10 \mid 200]$$

$$R_3 \rightarrow (1/10)R_3 \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 1 \mid 20]$$

$$x_3 = 20, x_2 = 80 - 2(20) = 40, x_1 = -40 + 4(20) = 40$$

Check: $40 + 40 + 20 = 100$ kg total, copper $20 + 8 + 8 = 36$ kg.

[ANSWER: 40 kg of Alloy 1, 40 kg of Alloy 2, 20 kg of Alloy 3]

Problem 5

(a)

$$[1 \ 2 \ 1]$$

$$[2 \ 4 \ 3]$$

$$[3 \ 6 \ 4]$$

$$R_2 \rightarrow R_2 - 2R_1 \quad [1 \ 2 \ 1]$$

$$R_3 \rightarrow R_3 - 3R_1 \quad [0 \ 0 \ 1]$$

$$[0 \ 0 \ 1]$$

$$R_3 \rightarrow R_3 - R_2 \quad [1 \ 2 \ 0]$$

$$R_1 \rightarrow R_1 - R_2 \quad [0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]$$

$$[\text{ANSWER: } \text{RREF}(A) = [1 \ 2 \ 0]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]]$$

(b)

$\text{rank}(A) = 2$, since the RREF has two pivots (columns 1 and 3) and two nonzero rows.

$$[\text{ANSWER: } \text{rank}(A) = 2]$$

(c)

0 solutions: possible.

1 solution: impossible.

Infinitely many: possible.

Problem 6

(a)

These are dependent because there are three vectors and v_3 has a zero entry, so they can't fill $\mathbb{R}^3$.

[ANSWER: dependent, span is a plane]

(b)

None of these is a multiple of another so they are independent.

[ANSWER: independent, span is $\mathbb{R}^3$]